

in blocks, enabling hundred per cent coverage. N1 can also recognise a carpet, and automatically lift its mop and switch to sweep only mode when moving over the carpet.

It has a multi-function app for remote control; for example, you can draw a specific area on the app for N1 to clean or enable N1 to switch to mop mode. You may also set up daily cleaning timetable for the robot through the app. Other options include auto cleaning, regular cleaning, fixed-point cleaning, specific-area cleaning, mode switching and one-click recharge.

N1's improved configuration makes cleaning more assured with 2000Pa strong suction power, easy-to-overcome 2cm obstacles without stress, 5200mAh large capacity lithium battery, two-hour-long battery life and auto-recharging when the battery is low. The bot resumes work from the same spot after recharging to full. Its collision sensor protects the machine and furniture. The ground detection sensor prevents it from falling when facing the stairs.

By attaching a dedicated PTZ, N1 becomes a moving selfie assistant at home. N1 app has an independ-

ent shooting function; you can set the camera movement route for N1 through the app. N1 will move around, holding the camera to help you shoot.

N1 has such additional features as continuous scanning, LCD display, dust recognition and laser navigation. It weighs seven kilograms, can cover an application area up to 400m², makes only 40dB-60dB noise, comes with twelve-month warranty, three to five hours charging time and four-core 1.3GHz Cortex A35 chip. It is priced around ₹ 1700.

—Deepshikha Shukla

5 SMART CONNECTED BATTERY SOLUTION FOR ELECTRIC VEHICLES

Right now, the world is in the middle of a technological paradigm shift, namely, renewable energy and electric vehicles (EVs). Being a new technology, there are not many players with prior experience of lithium-ion batteries.

Apart from this, all other challenges related to taking new technology to market are present. For example, finding the right supplier of equipment, developing new manufacturing techniques to optimise production, coming up with efficient and safe standard operating procedures, and so on.

One specific problem is about safety from high-voltage DC electric shock. The industry currently does not have much expertise in working with very-high-voltage DC in the range of 800V DC to 1000V DC. So, a lot of safety training is needed to avoid mishaps.

Pain points of EVs are sub-par performance in Indian terrain and temperature, lithium-ion battery for high power vehicles, interference in connectivity/communication, thermal management, safety requirements at high voltage and current, fleet operators' issue with starter battery, need for fast charging, need for communication and control over starter battery operation.

To ensure India plays a crucial role in this changeover and remove

dependency on other countries for its EV components supply, IIT-Madras incubated deep-tech startup Grinntech has designed lithium-ion batteries that can be used for automotive applications and stationary energy-storage applications. Smart energy storage systems will be the foundation of how India will use energy in the future.

Lithium-ion batteries are very application- and environment-specific. So, it is not possible to design batteries without customising them for intended applications and environments. Grinntech solves this problem by custom-designing smart connected battery packs for specific applications.

Properties that make these batteries unique are analysis and simulation of cell characterisation, thermal simulation, variation and shock simulation, battery management system design, protection functionalities, mechanical design and packaging, cooling system design, current-carrying parts design, embedded software for monitoring and control, estimation algorithms, application business logic, communication protocol, protection against excessive current, protection against high voltage or leakage current, and noise/interference isolation between power and communication.

With inbuilt, compact liquid

cooling, only four per cent battery volume is used for cooling. It has fast charging design with three times faster charging, is lockable via communication predictive troubleshooting and maintenance, and features connected battery, high current applications and ultra-low-power sleep mode.

Nikhilesh Mishra, chief executive officer, Grinntech Motors & Services Pvt Ltd, explains, "There have been many tests to ensure performance of batteries and vehicles throughout the year to ensure performance in all terrains and temperatures. Automotive OEMs are the primary target customer. Although, most of the EV business currently is being done in B2B segment where vehicles are being used for such applications as ride-sharing, food delivery, courier delivery and so on."

In general, the EV experience is better than petrol or diesel for similarly power-rated vehicles. Pick-up is better, rides are noise-free and there is not much vibration.

There have been concerns about battery life and performance that Grinntech is trying to solve. Vehicles with Grinntech batteries are giving more range and faster charging. Battery degradation is low, and users know accurately how much energy is left in the batteries so they are never stranded on the road. **EFY**

—Deepshikha Shukla